GO Platform v7.0 — Architecture Diagrams

> Architecture diagrams for the Hyperledger Fabric Guarantee of Origin platform. Updated for v7.0 (cross-registry bridge, IoT device attestation, external data oracle). Diagrams use Mermaid syntax and render natively on GitHub.

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1. System Architecture Overview

Shows the full three-tier stack: React frontend → Express backend → Hyperledger Fabric network with tiered organizations.

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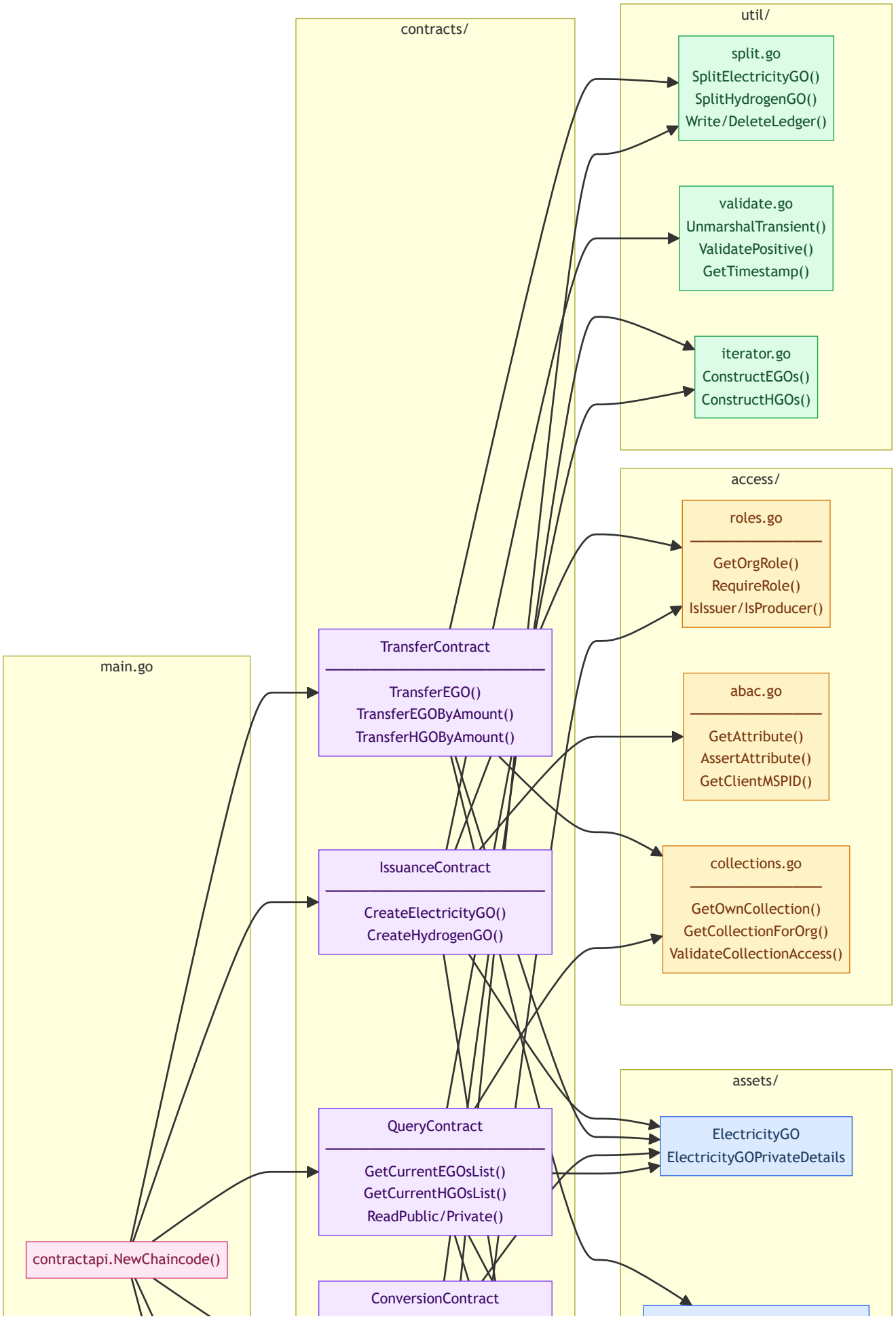
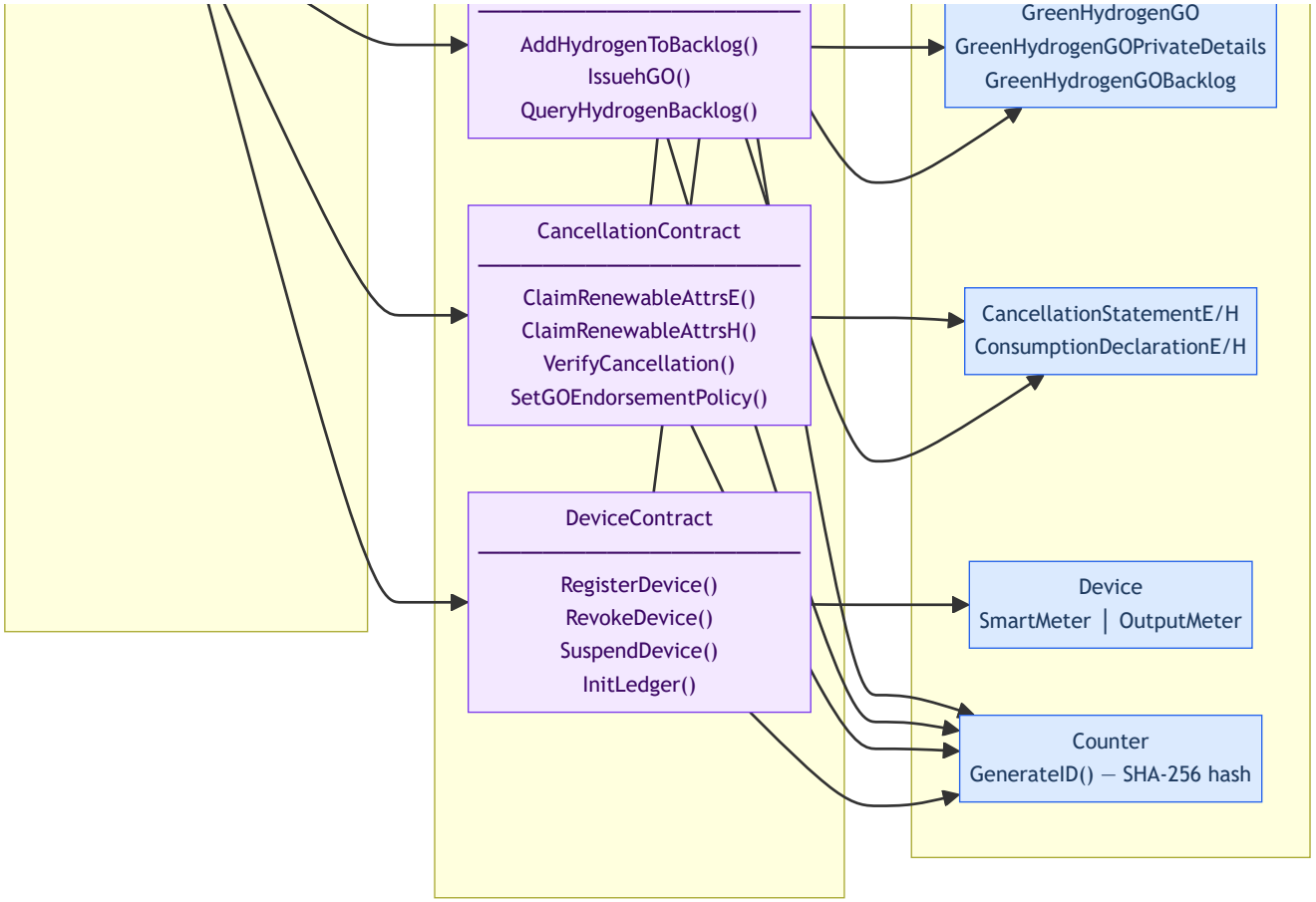
2. Chaincode Internal Architecture — Package Dependencies

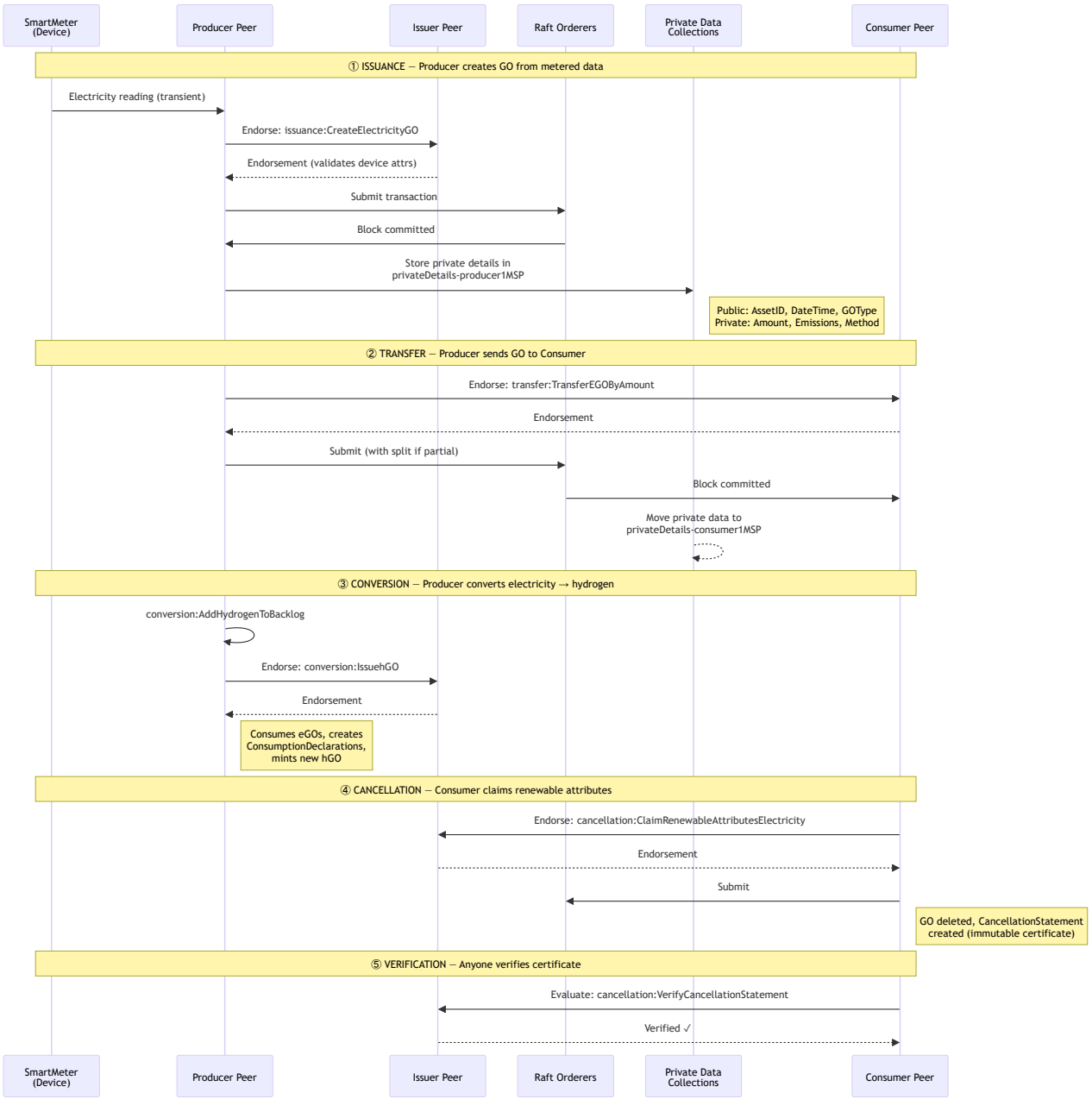
Shows how the 6 named contracts in `contracts/` depend on shared packages: `assets/`, `access/`, and `util/`.


contracts/
util/
split.go
SplitElectricityGO()
SplitHydrogenGO()
Write/DeleteLedger()
validate.go
UnmarshalTransient()
ValidatePositive()
GetTimestamp()
iterator.go
ConstructEGOs()
ConstructHGOs()
access/
roles.go
GetOrgRole()
RequireRole()
IsIssuer/IsProducer()
abac.go
GetAttribute()
AssertAttribute()
GetClientMSPID()
collections.go
GetOwnCollection()
GetCollectionForOrg()
ValidateCollectionAccess()
main.go
TransferContract
TransferEGO()
TransferEGOByAmount()
TransferHGOByAmount()
IssuanceContract
CreateElectricityGO()
CreateHydrogenGO()
QueryContract
GetCurrentEGOsList()
GetCurrentHGOsList()
ReadPublic/Private()
ConversionContract
assets/
ElectricityGO
ElectricityGOPrivateDetails
contractapi.NewChaincode()

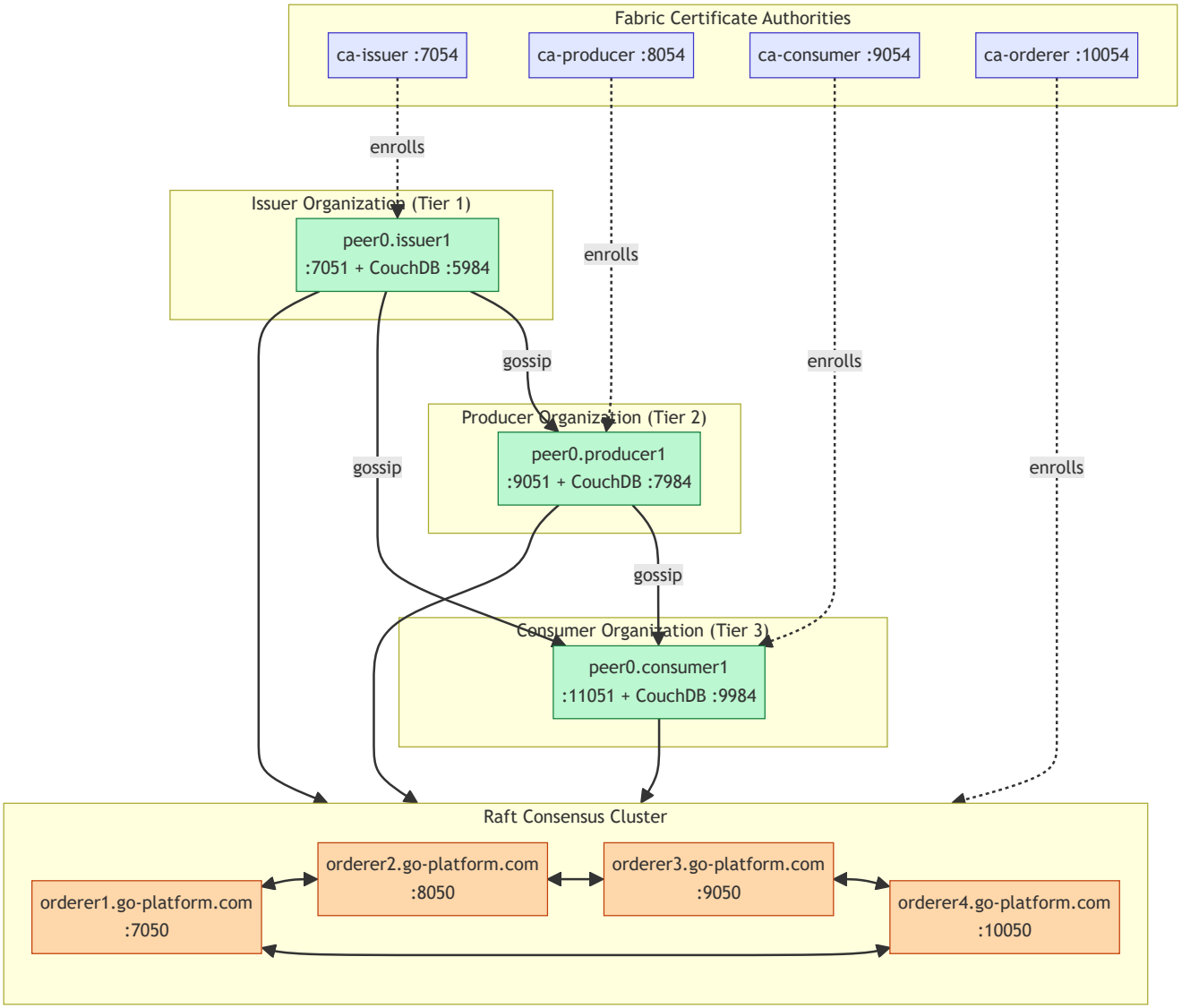


3. GO Lifecycle Sequence — Issuance → Transfer → Conversion → Cancellation → Verification

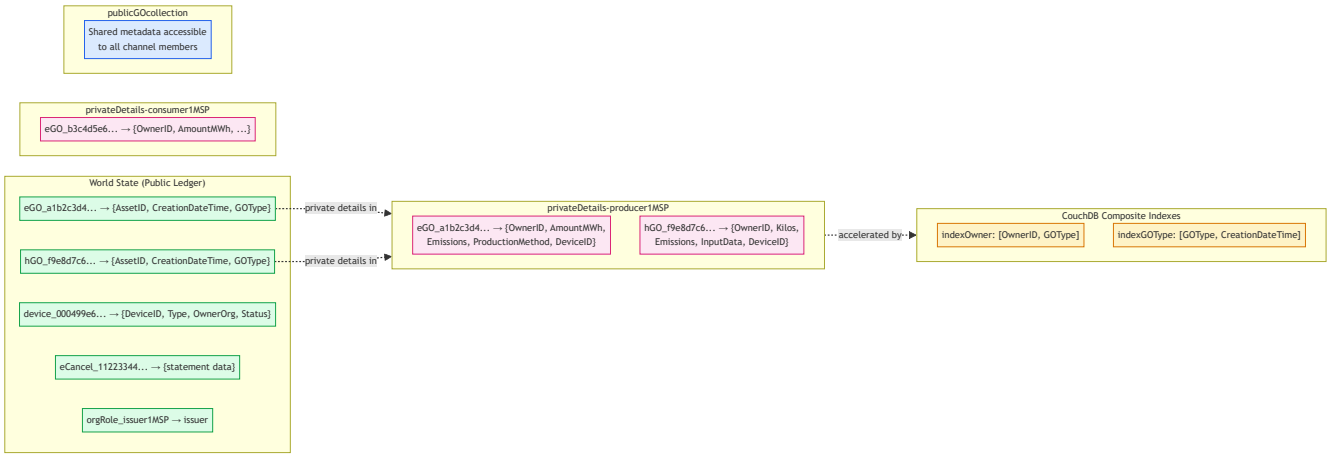
Shows the end-to-end flow of a Guarantee of Origin through the system, from metered data to verified certificate.


SmartMeter (Device)
Producer Peer
Issuer Peer
Raft Orderers
Private Data Collections
Consumer Peer
① ISSUANCE — Producer creates GO from metered data
Electricity reading (transient)
Endorse: issuance:CreateElectricityGO
Endorsement (validates device attrs)
Submit transaction
Block committed
Store private details in privateDetails-producer1MSP
Public: AssetID, DateTime, GOType
Private: Amount, Emissions, Method
② TRANSFER — Producer sends GO to Consumer
Endorse: transfer:TransferEGOByAmount
Endorsement
Submit (with split if partial)
Block committed
Move private data to privateDetails-consumer1MSP
③ CONVERSION — Producer converts electricity → hydrogen
conversion:AddHydrogenToBacklog
Endorse: conversion:IssuehGO
Endorsement
Consumes eGOs, creates ConsumptionDeclarations, mints new hGO
④ CANCELLATION — Consumer claims renewable attributes
Endorse: cancellation:ClaimRenewableAttributesElectricity
Endorsement
Submit
GO deleted, CancellationStatement created (immutable certificate)
⑤ VERIFICATION — Anyone verifies certificate
Evaluate: cancellation:VerifyCancellationStatement
Verified ✓
SmartMeter (Device)
Producer Peer
Issuer Peer
Raft Orderers
Private Data Collections
Consumer Peer

4. Network Topology

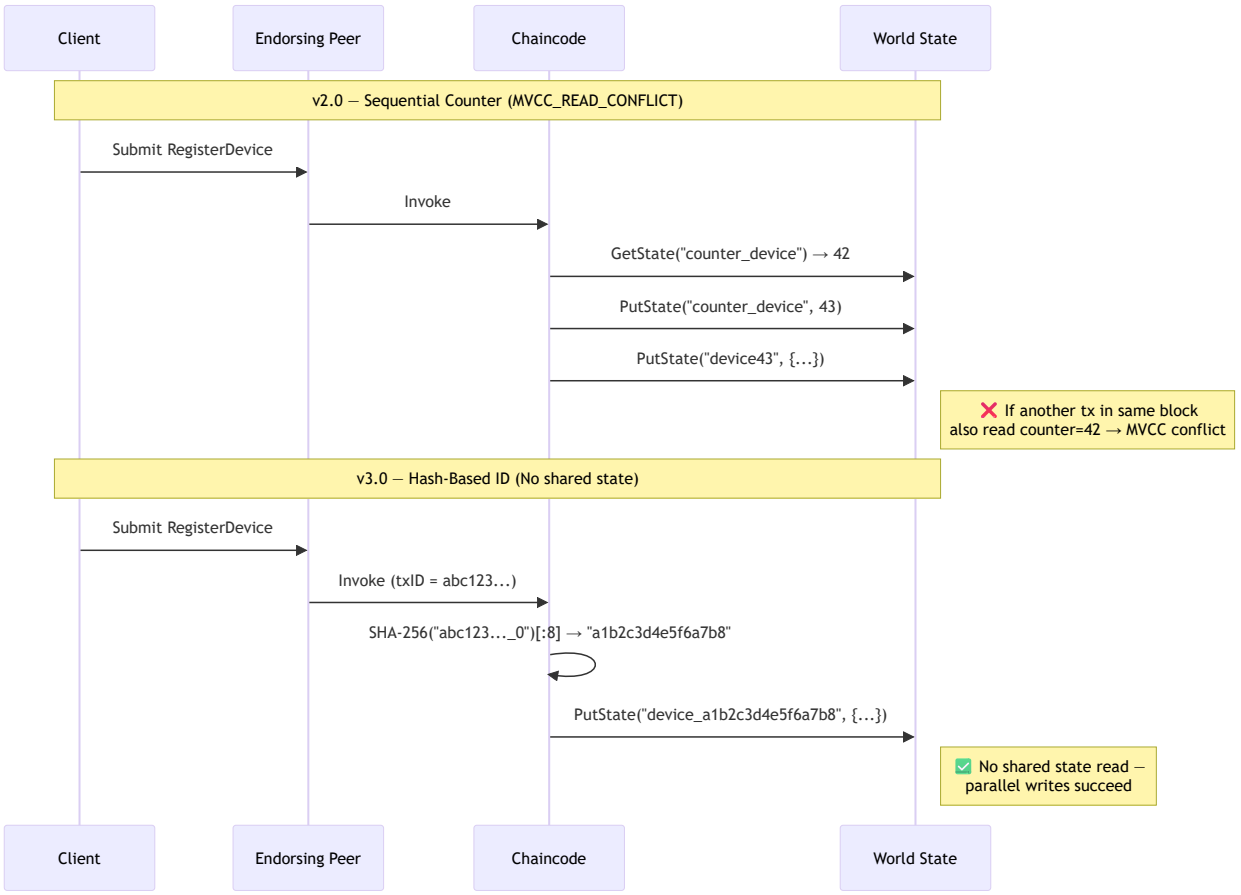


5. Data Architecture — Public vs. Private State



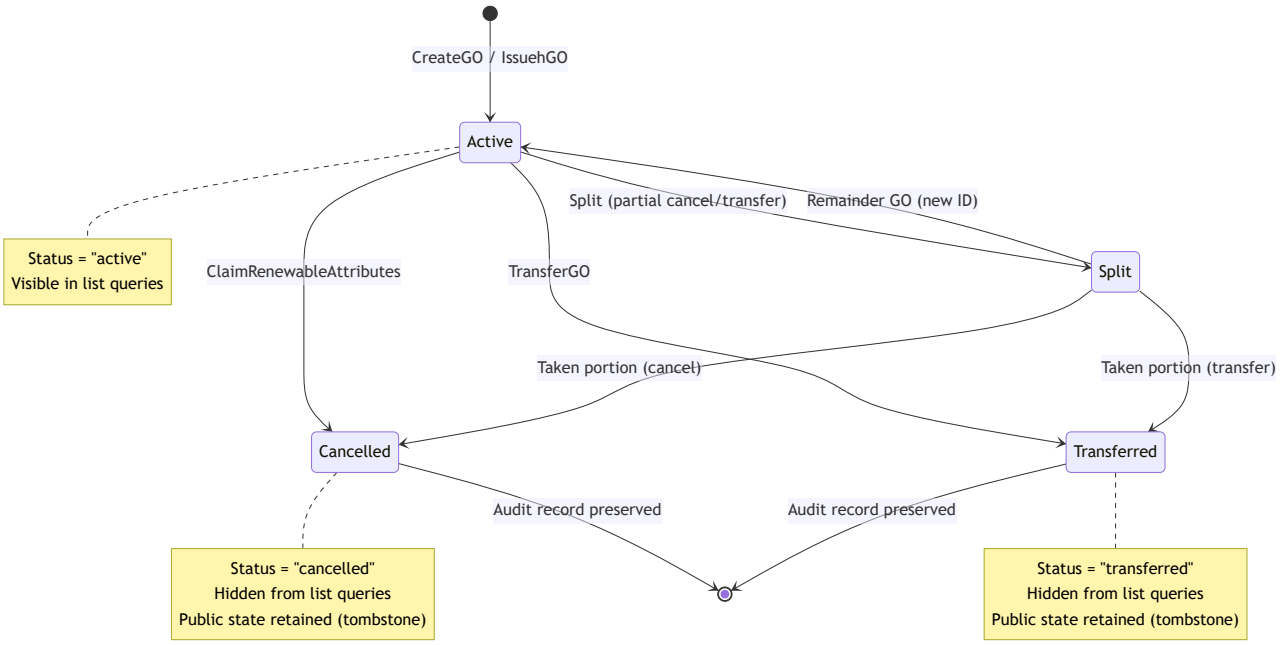
6. v3.0 ID Generation Flow

Shows how v3.0 generates contention-free IDs from the transaction ID, eliminating the MVCC_READ_CONFLICT bottleneck.



7. v4.0 GO Lifecycle State Machine (ADR-007 Tombstone)

Shows the lifecycle states introduced by the tombstone pattern.



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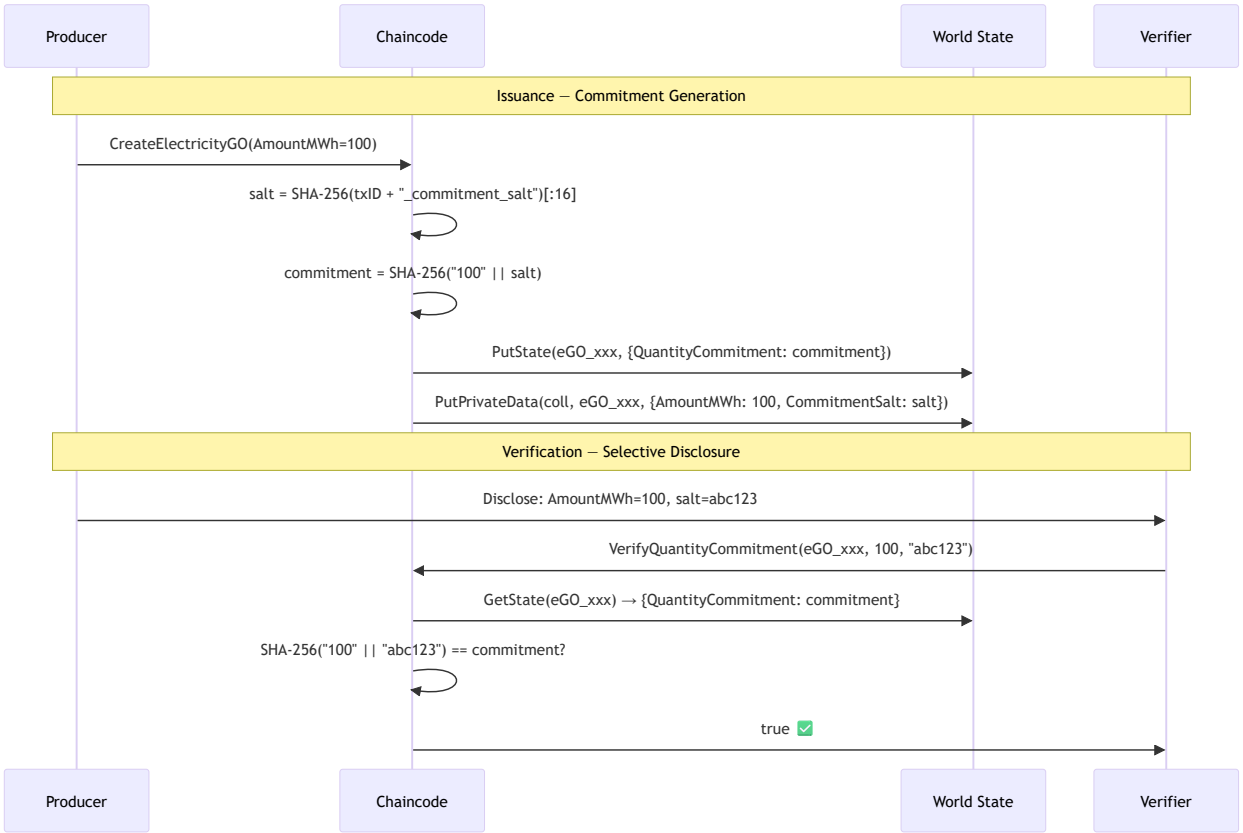
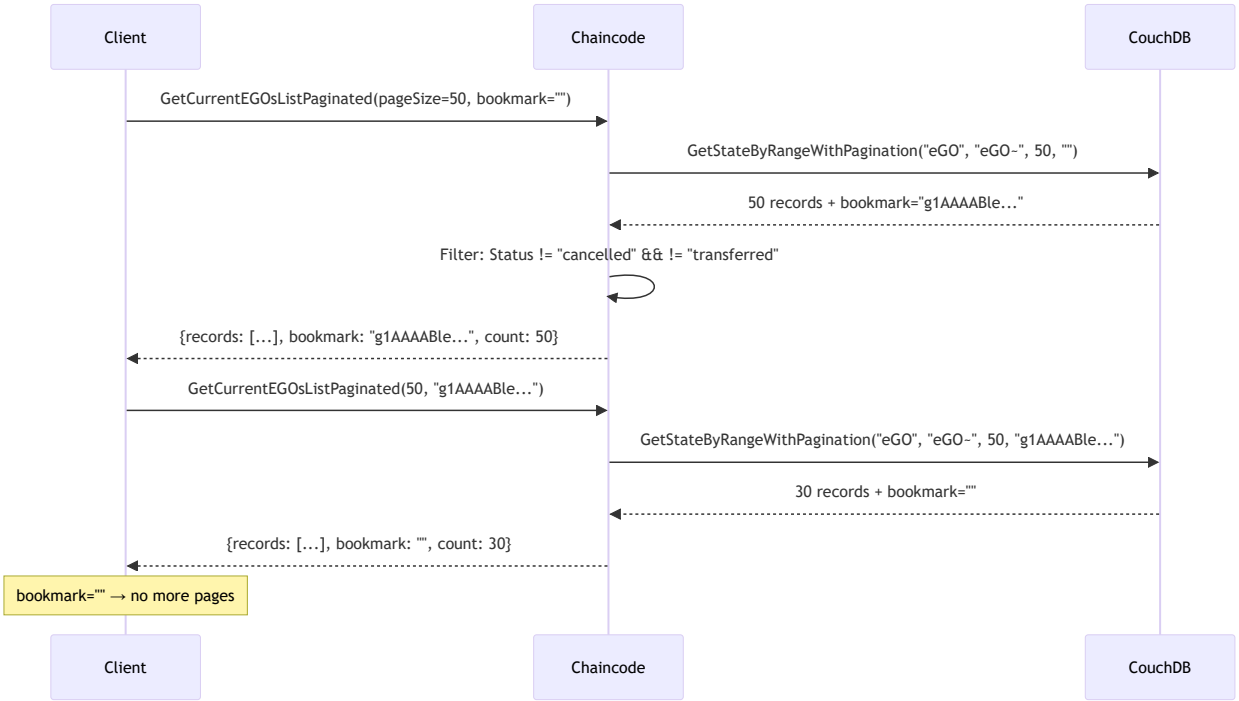
8. v4.0 Selective Disclosure (ADR-009)

Shows the hash commitment scheme enabling quantity verification without collection access.

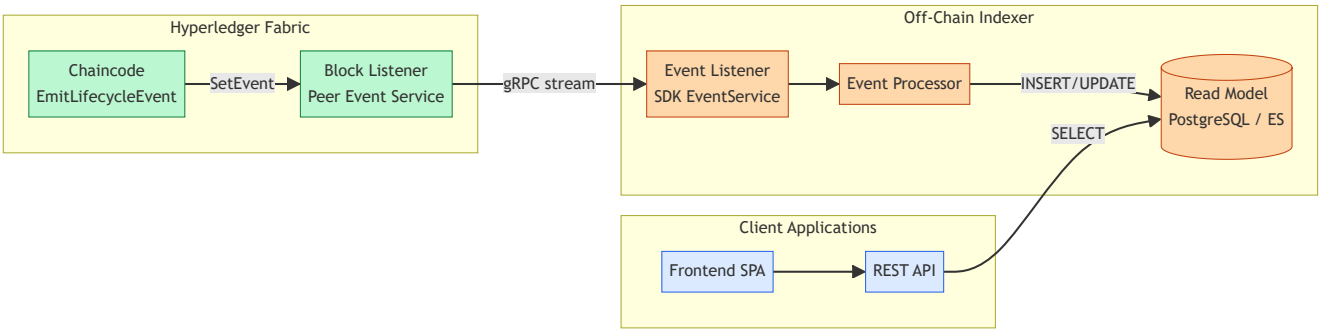


9. v4.0 Paginated Query Flow (ADR-006)



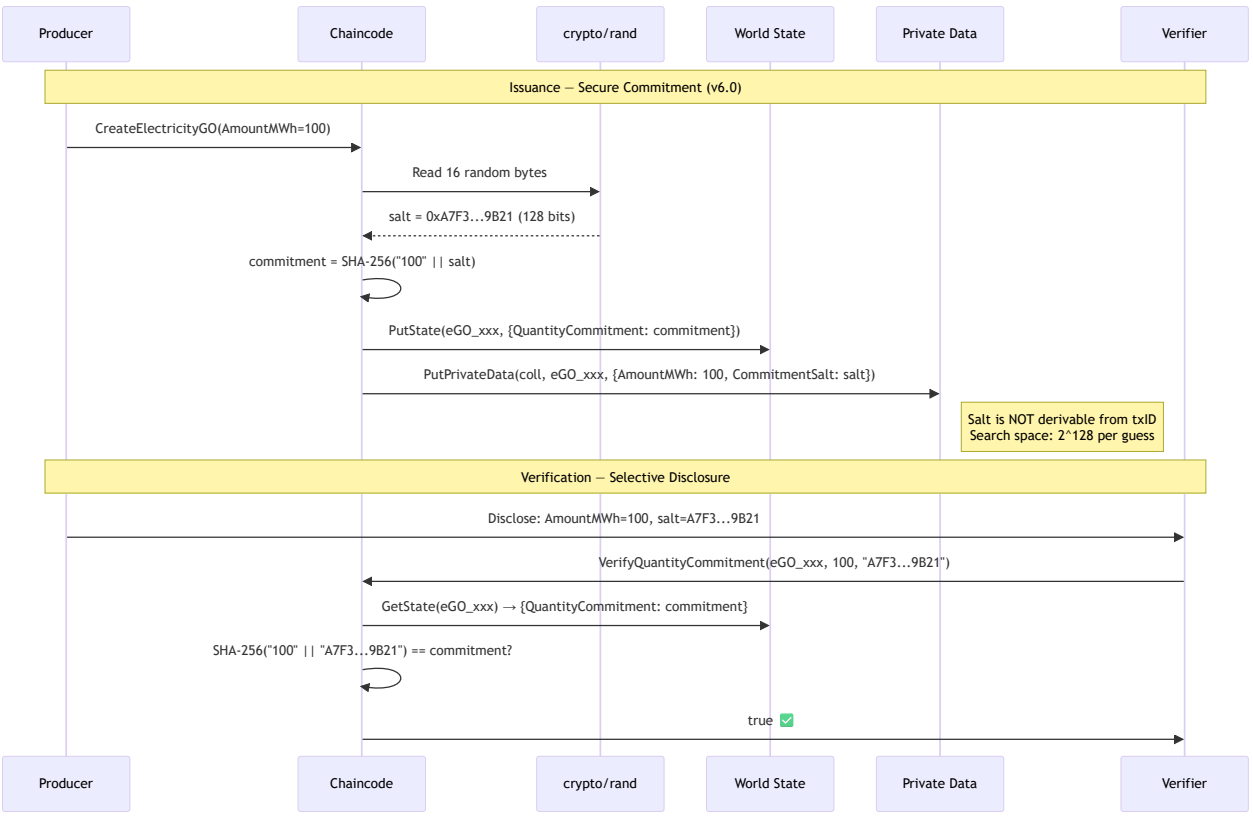
10. v7.0 Contract Architecture (10 Namespaces)

11. v5.0 CQRS Event Flow (ADR-016)



12. v6.0 Selective Disclosure with Secure Salts (ADR-009 + ADR-017)

Shows the improved hash commitment scheme using `crypto/rand` instead of `txID`-derived salts.



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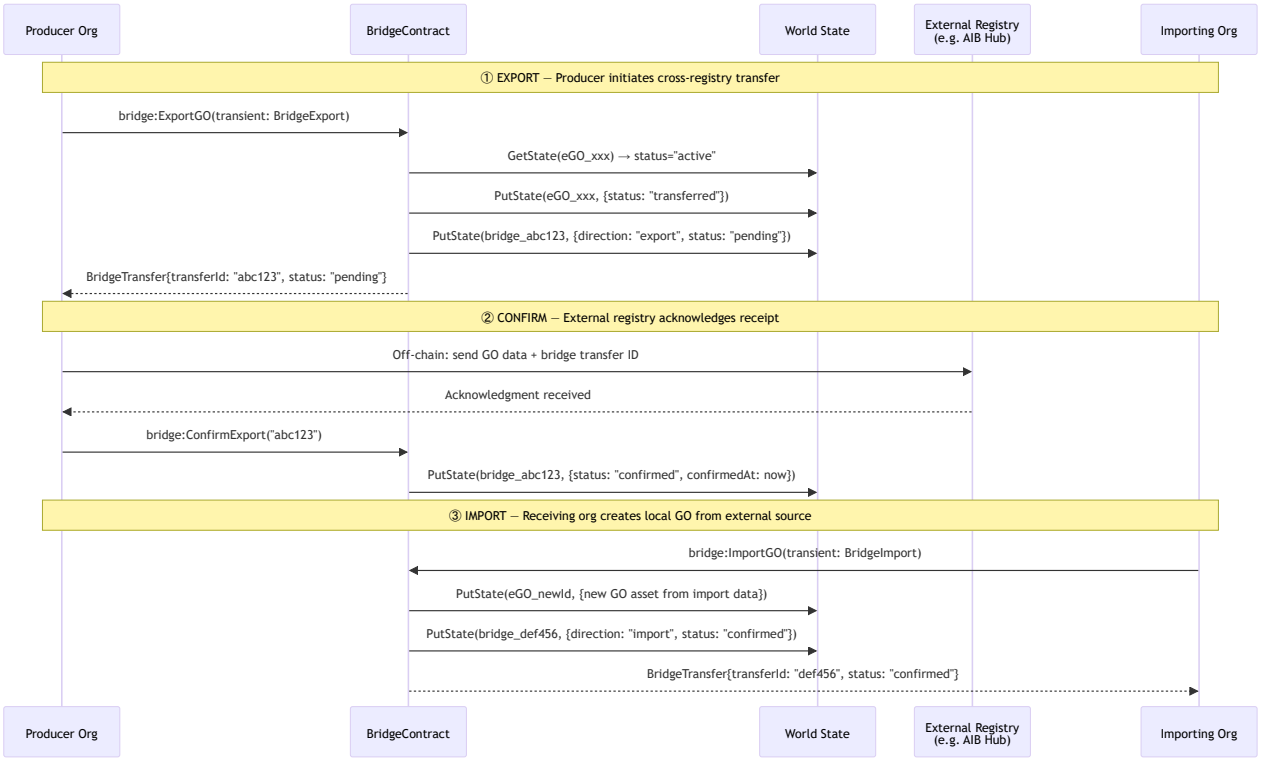
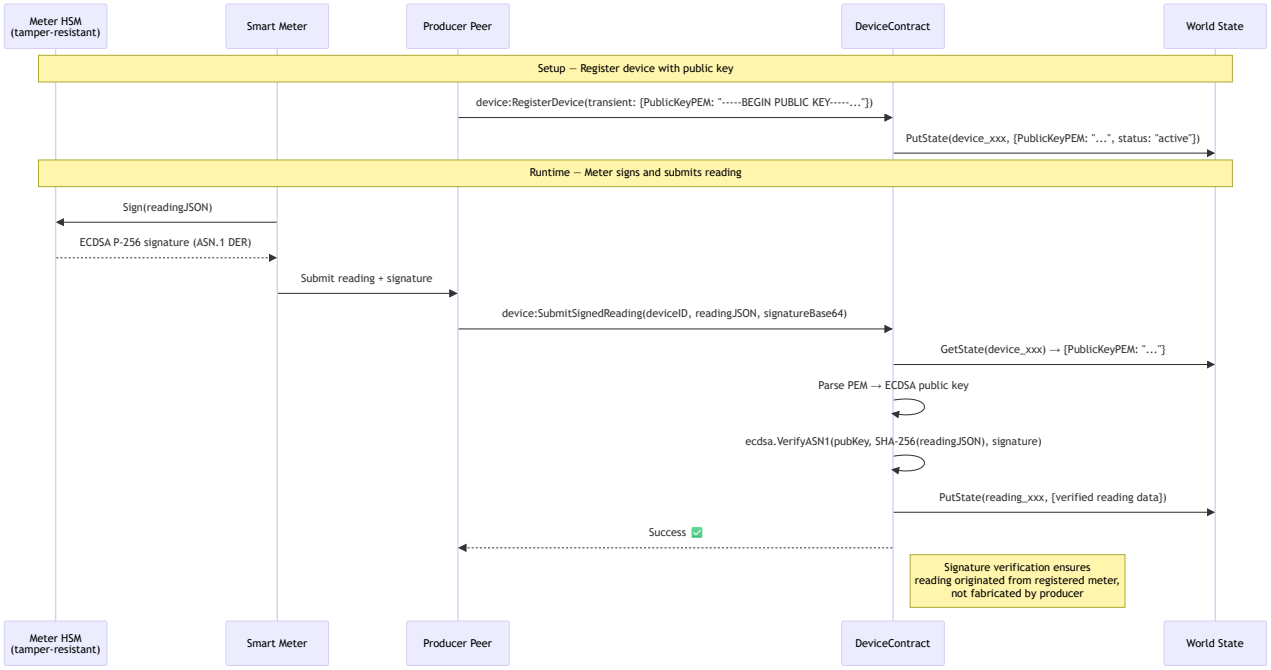
13. v7.0 Cross-Registry Bridge Flow (ADR-024)

Shows the three-phase bridge protocol for GO export and import between registries.



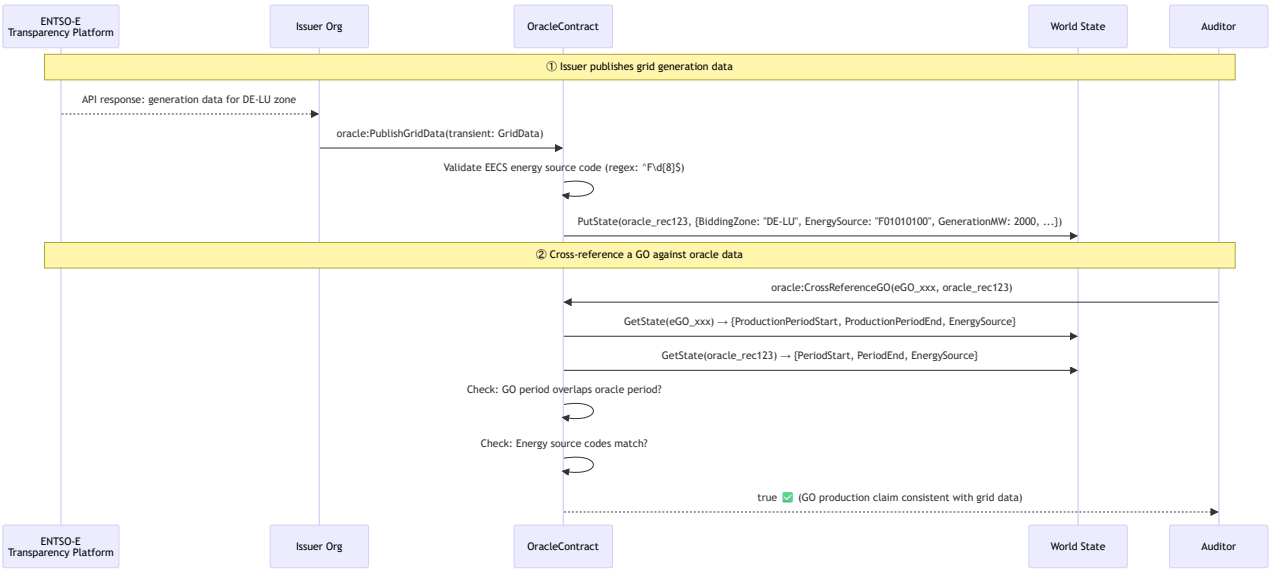
14. v7.0 IoT Device Attestation Flow (ADR-027)

Shows how physical smart meters cryptographically attest their readings on-chain.



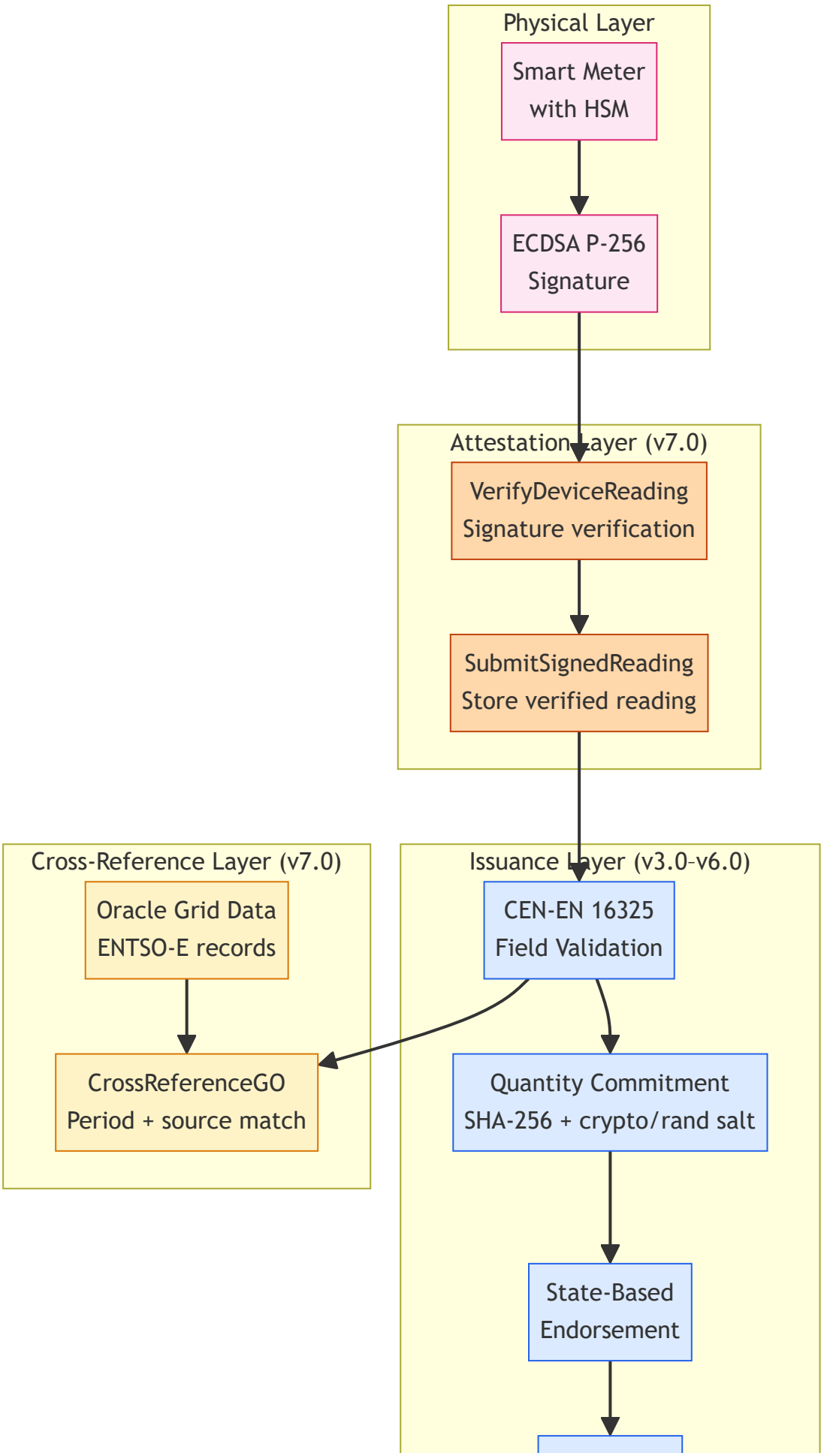
15. v7.0 Oracle Cross-Reference Flow (ADR-029)

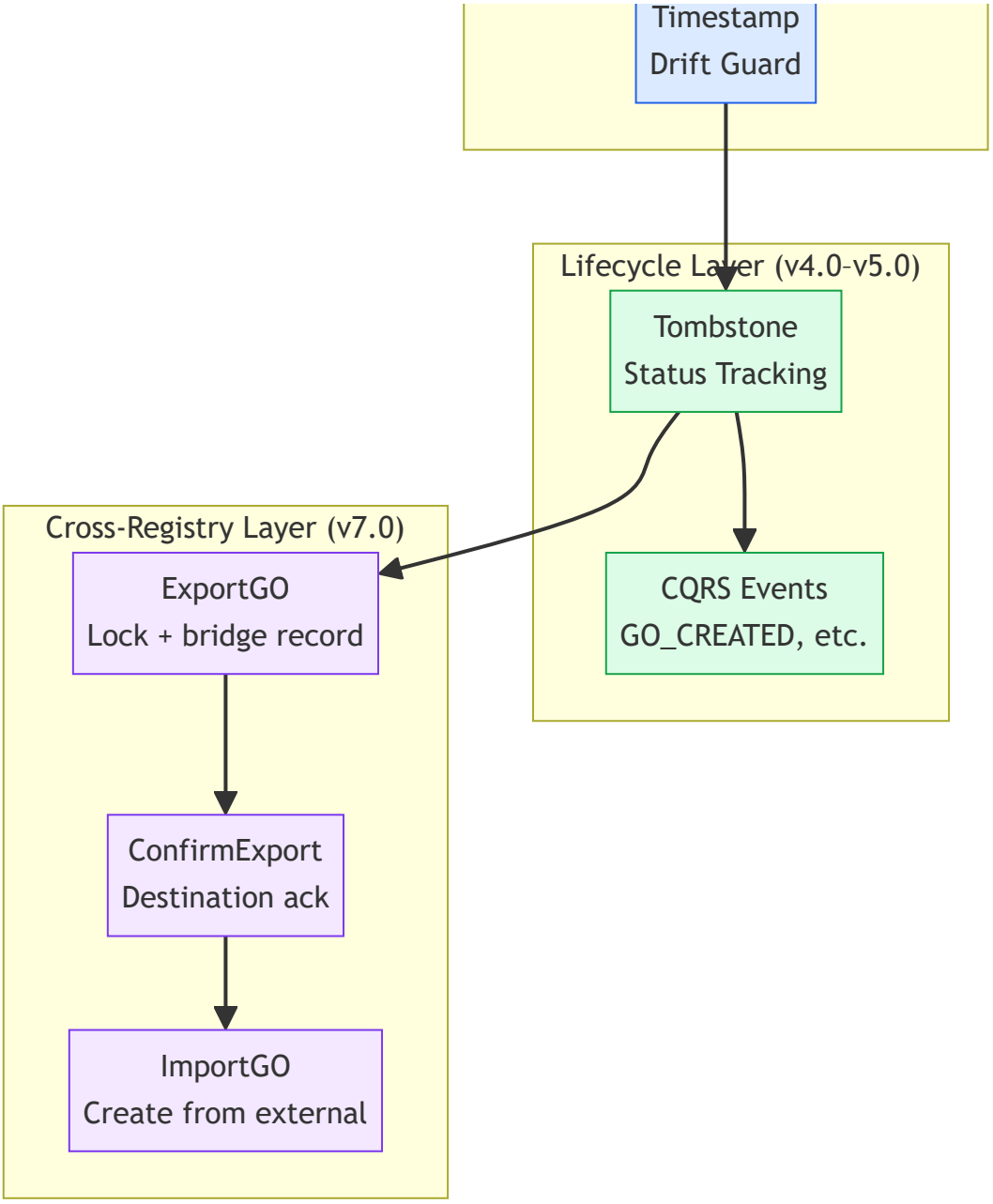
Shows how GO production claims are validated against external grid generation data.



16. v7.0 Verifiability Stack — End-to-End Provenance

Shows how v7.0 completes the full provenance chain from physical meter to cross-border transfer.


Physical Layer
Smart Meter with HSM
ECDSA P-256 Signature
Attestation Layer (v7.0)
VerifyDeviceReading
Signature verification
SubmitSignedReading
Store verified reading
Cross-Reference Layer (v7.0)
Oracle Grid Data
ENTSO-E records
CrossReferenceGO
Period + source match
Issuance Layer (v3.0–v6.0)
CEN-EN 16325
Field Validation
Quantity Commitment
SHA-256 + crypto/rand salt
State-Based
Endorsement



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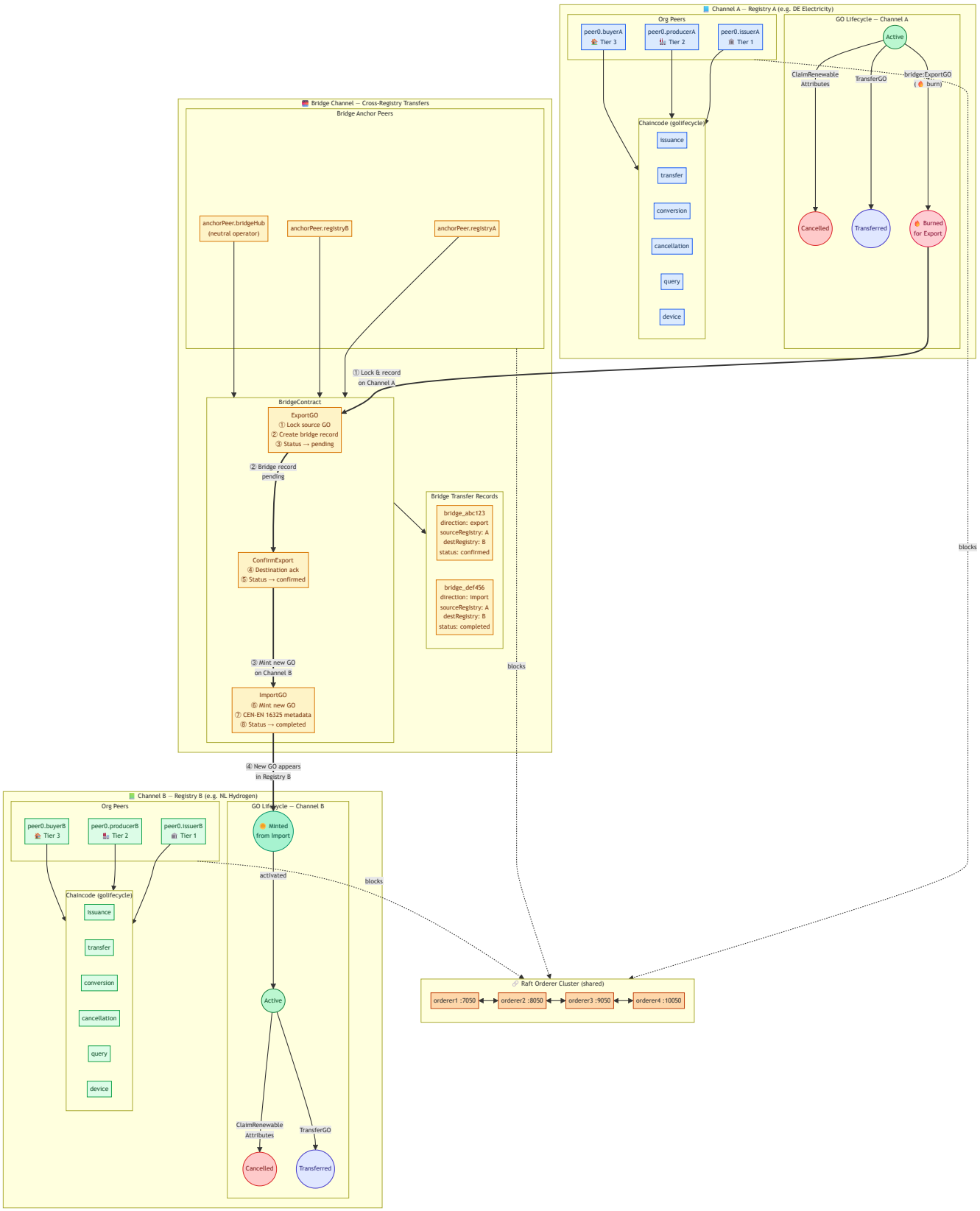
17. v7.0 Multi-Channel Architecture with Bridge Burn-and-Mint

Combines the network topology, the GO lifecycle state machine, and the multi-channel bridge architecture into one unified diagram. Shows how GOs flow through lifecycle states within national registry channels, while the bridge channel coordinates cross-registry transfers via a burn-and-mint pattern.


Channel A — Registry A (e.g. DE Electricity)
Org Peers
peer0.buyerA
Tier 3
peer0.producerA
Tier 2
peer0.issuerA
Tier 1
Chaincode (golifecycle)
issuance
transfer
conversion
cancellation
query
device
GO Lifecycle — Channel A
Active
ClaimRenewable Attributes
TransferGO
bridge:ExportGO (burn)
Cancelled
Transferred
Burned for Export
Bridge Channel — Cross-Registry Transfers
Bridge Anchor Peers
anchorPeer.bridgeHub (neutral operator)
anchorPeer.registryB
anchorPeer.registryA
① Lock B record on Channel A
BridgeContract
ExportGO
① Lock source GO
② Create bridge record
③ Status → pending
② Bridge record pending
ConfirmExport
④ Destination ack
⑤ Status → confirmed
③ Mint new GO on Channel B
ImportGO
⑥ Mint new GO
⑦ CEN-EN 16325 metadata
⑧ Status → completed
Bridge Transfer Records
bridge_abc123
direction: export
sourceRegistry: A
destRegistry: B
status: confirmed
bridge_def456
direction: import
sourceRegistry: A
destRegistry: B
status: completed
blocks
blocks
④ New GO appears in Registry B
Channel B — Registry B (e.g. NL Hydrogen)
Org Peers
peer0.buyerB
Tier 3
peer0.producerB
Tier 2
peer0.issuerB
Tier 1
Chaincode (golifecycle)
issuance
transfer
conversion
cancellation
query
device
GO Lifecycle — Channel B
Minted from Import
activated
Active
ClaimRenewable Attributes
TransferGO
Cancelled
Transferred
blocks
Raft Orderer Cluster (shared)
orderer1 :7050
orderer2 :8050
orderer3 :9050
orderer4 :10050